



HK IMO Training 2021-2022
Problem Set 19
For 26 February 2022



Problem 73. Find the largest positive integer n which cannot be expressed as the sum of three pairwise coprime integers larger than 1.

Problem 74. In $\triangle ABC$, $\angle C = 90^\circ$. Let M be the midpoint of AB . The incircle of $\triangle BCM$ touches BC and BM at P and Q respectively. The incircle of $\triangle ACM$ touches AC and AM at S and T respectively. Prove that PQ , ST and the internal angle bisector of $\angle ACB$ are concurrent.

Problem 75. Do there exist polynomials P, Q with integer coefficients such that $\deg P, \deg Q \geq 2022$ and

$$P(Q(x)) = 3Q(P(x)) + 1$$

for all real numbers x ?

Problem 76. There is a tournament of $n \geq 7$ players such that every pair of players may play against each other for an arbitrary number of times (possibly 0 times). It is possible that player A has won over player B in one game, and B has won over A in another game. It is known that for every two players, there is a third player who has won over each of them. Prove that for any positive integer k satisfying $2(2^{2^k} - 1) \geq n$, there exists $m \in \{2, 3, \dots, 2k\}$ and players $A_1, A_2, \dots, A_m$ such that A_{i+1} has won over A_i for $i = 1, 2, \dots, m$, where $A_{m+1} = A_1$.